

EDUCATION

Cornell University, Ithaca, NY

PhD in Plant Breeding and Genetics, August 2026 (expected)

Concentration: Plant Breeding and Computational Genetics, minor in Geospatial Applications

Thesis: “Genomic and Phenomic Approaches for Barley Breeding for Malting Quality and Abiotic Stress Resistance”

Responsibilities: Administering a malting barley breeding program to breed varieties for NY under Prof. M.E Sorrells

Skills: UAV Pilot, Statistical Modelling, QGIS, Python, R, Genomic Prediction, GWAS, Novel Phenotyping Approaches

Activities: Graduate and Professional Student Assembly, *Biological Sciences Representative, VP of Finance*

Swedish University of Agricultural Sciences, Uppsala, Sweden

M.Sc. in Plant Biology, June 2022 GPA: 5.00/5.00

Thesis: “Identification of New Loci and Candidate Genes for Resistance to Sclerotinia Stem Rot in *B. napus* with GWAS”

Estonian University of Life Sciences, Tartu, Estonia

M.Sc. in Agriculture, June 2021 GPA: 4.90/5.00, cum laude

Thesis: “Adaption of Estonian Grown Barley Varieties to Climate Change Induced Water Regime Changes”

Awards: Kristjan Jaak Scholarship for Exchange Studies

Estonian University of Life Sciences, Tartu, Estonia

B.Sc. in Agriculture June 2019 GPA: 4.74/5.00, cum laude

Thesis: “Wheat Flag Leaf Wheat in Organic and Conventional Systems and the Effect on Yield and Quality”

Activities: University Assembly, *Student Representative*

Department of Academic Affairs, *International Student Tutor*

Student Association of Young Agricultural Enthusiasts of Estonia, *President, VP*

Estonian Student Association of Biosciences, *Science Communication Team Member*

Awards: Organic Farming Research Centre Scholarship, Veski Mati Scholarship, Rotalia Foundation Scholarship, ERASMUS for Young Entrepreneurs Scholarship (Bioinnovation Hackathon)

Exchanges: University of Aix-Marseille, FRA, University of Kaunas, LIT, Summer School: Soil and Water

EXPERIENCE

Estonian Crop Research Institute, Jõgeva, Estonia

Senior Research Agronomist, June 2020 – Aug 2022

Managed phenotyping and supported the management in the barley breeding program. Ran HTP crop canopy phenotyping experiments for climate resilience, providing data for NoBalWheat and SusCcrop projects.

Estonian Ministry of Agriculture and Rural Life, Tallinn, Estonia

Development Team Member, Sep 2018 – Nov 2018

Participated in a team drafting a development and innovation plan (Agricultural Crops 2030) for the national government. Researched, presented, and led the cereal sector innovation discussion group.

Ulm University, Kazda lab, Ulm, Germany

Lab Intern, June – Aug 2018

Operated biogas-reactors running on silage. Ran routine chemical analysis for the lab bioreactors and local biogas plants.

PUBLICATIONS

- **Sepp S. S.**, Tamm Ü., Loit E. 2021. Is Estonian Barley Ready to Tackle Climate Change Induced Water Regimes? Agraarteadus (Journal of Agricultural Science). <https://doi.org/10.15159/jas.21.29>
- Chawade A., Thuraga V., **Sepp S. S.**, Ghosh S., Ghadamgahi F., Odilbekov F., Muthusamy S., Vetukuri R. R., Abreha K. B. 2025. Genomic loci for sclerotinia stem rot resistance and chlorophyll stability in Brassica napus: integrating GWAS with microbiome insights. Journal of Plant-Environment Interactions. <https://doi.org/10.1002/pei3.70092>
- Kolkman J. M., **Sepp S. S.**, Kunze K. H., Bergstrom G. C., Sorrells M. E. 2026. Genome-wide association analysis of resistance to scald in an adapted multiparent winter malting barley population. bioRxiv 2026.03.12.711358
- **Sepp S.S.**, Kunze K.H., Bergstrom G.C., Kolkman J.M., Hayes P., Sorrells M. E. 2025. Winter malting barley LakeEffect. Journal of Plant Registration. (*In review*)